Real-World Use of PCSK9 Inhibitors and Adherence to Guideline-Directed Therapy in an Integrated Specialty Pharmacy

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BACKGROUND

- Proprotein Convertase Subtilisin/Kexin Type 9 inhibitors (PCSK9i), evolocumab and alirocumab, decrease low-density lipoprotein (LDL) levels and risk of coronary events.[1,2]
- Clinical guidelines recommend PCSK9i therapy for the following.[3]

Patients on maximally tolerated statin therapy and ezetimibe with

- Atherosclerotic cardiovascular disease (ASCVD) LDL ≥70mg/dL
- Familial hypercholesterolemia with LDL ≥100mg/dL

- Research evaluating adherence to these guidelines in real-world clinical practice has been scarce.

OBJECTIVES

- To assess adherence to guideline-directed lipid management and PCSK9i use in a real-world setting
- To identify clinical outcomes after 24 months of PCSK9i therapy

METHODS

Design	Single-center retrospective review conducted at Vanderbilt University Medical Center (VUMC)
Inclusion	Patients prescribed a PCSK9i by a VUMC provider from September 2015 to August 2018
Exclusion	• Received a PCSK9i through a manufacturer patient assistance program (PAP) • Never received insurance approval • Never started therapy
Outcomes	• Adherence to guideline-directed lipid-lowering therapy • Attainment of LDL goal based on indication (≤ 70mg/dL for ASCVD or ≤ 100mg/dL for heterozygous familial hypercholesterolemia [HeFH]) • Change in LDL from baseline to 24 months • Incidence of adverse cardiovascular events or new onset diagnosis after PCSK9i initiation

RESULTS

TABLE 1. SAMPLE CHARACTERISTICS (N=477)

Characteristics	% (n) or Median [IQR]
Age, years	63 [56-70]
Race	
White	91% (436)
Black	7% (33)
Other/Unknown	2% (8)
Gender, Male	53% (253)
Medical history	
Hypertension	86% (408)
Diabetes	35% (168)
Tobacco user	11% (52)
Historical lipid management	(477)
Ezetimibe	49% (235)
Statin	98% (467)
None	2% (9)
LDL, Baseline	157 [131-190]

TABLE 2. CLINICAL TREATMENT INDICATIONS FOR INITIAL PCSK9I THERAPY (N=477)

PCSK9i Indication	% (n)
Familial hypercholesterolemia	41% (196)
Primary hyperlipidemia	3% (14)
ASCVD	86% (409)
ASCVD type *(n=409)*	
Acute coronary syndrome	13% (53)
Coronary revascularization Procedure (PCI, CABG)	72% (294)
Evidence of ASCVD via imaging studies	72% (295)
Myocardial infarction	42% (174)
Peripheral arterial disease	19% (78)
Positive coronary calcium score	13% (54)
Stable/unstable angina	59% (242)
Stroke	14% (56)
Transient ischemic attack	10% (41)

PCI= Percutaneous coronary intervention
CABG= Coronary artery bypass graft

FIGURE 1. PATIENT LIPID MANAGEMENT DURING AND AFTER INITIAL PSCK9I THERAPY

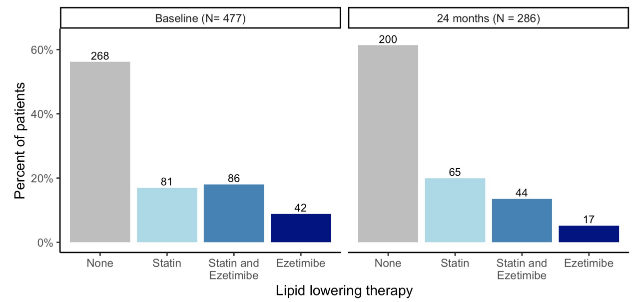
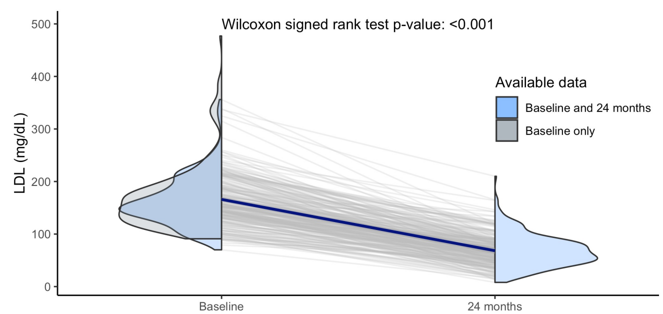
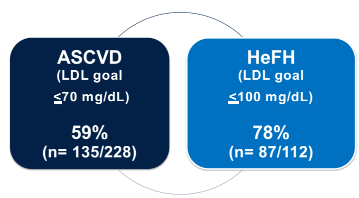


FIGURE 2. CHANGE IN LDL OVER 24 MONTHS OF PCSK9I THERAPY



Of the 262 patients with both baseline and 24-month LDL data, median LDL decreased from 157 mg/dL to 65 mg/dL (i.e. reduced by 59%)

FIGURE 3. LDL GOALS ACHIEVED AT 24 MONTHS (IN PATIENTS WITH AVAILABLE DATA)



67 patients incurred a total of 96 new onset cardiovascular events or diagnoses post-PCSK9i initiation.

TABLE 3. NUMBER OF NEW ONSET CARDIOVASCULAR EVENTS OR DIAGNOSES PER PATIENT

Number of Events or Diagnosis	% (n)
0	79% (259)
1	14% (44)
2	6% (18)
3+	1% (5)

CONCLUSIONS

- Less than half of patients were utilizing statin or ezetimibe therapy at the time of PCSK9i initiation as recommended by guidelines.
- Consistent with other clinical trials, our real-world study showed that most patients who were persistent to therapy at 24 months achieved their target LDL goal.
- Additional research is needed to evaluate PCSK9i use outside of guideline directed recommendations to fully evaluate the effects on future cardiovascular events.

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